

Deep Aman Ganta

AI Engineer

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PROFESSIONAL SUMMARY

- AI Engineer with close to 3 years of experience delivering full-cycle machine learning and AI solutions across diverse domains like automotive, e-commerce, and smart agriculture.
- Adept at building intelligent systems including fraud detection models, RAG-powered conversational agents, and AI-driven diagnostic tools using Lang Chain, Hugging Face, and OpenAI GPT.
- Dual-certified as an AWS Machine Learning Engineer Associate and Azure AI Engineer Associate, with deep expertise in constructing scalable pipelines using Sage Maker, Azure Cognitive Services, Pinecone, and Airflow.
- Strong foundation in machine learning and deep learning, with hands-on experience in NLP, transformers, and context-aware architectures tailored for real-world use cases.
- Advanced proficiency in implementing techniques such as clustering, segmentation, time series forecasting, and prompt engineering to optimize system intelligence and adaptability.
- Hands-on experience with MLOps tools including Docker, Kubernetes, and MLflow for automating deployment, monitoring, and governance of AI models in production.
- Skilled in developing cloud-native AI systems using RESTful APIs, microservices, and vector databases across AWS and Azure platforms to achieve performance and scalability goals.
- Committed to solving complex problems with innovative AI strategies that drive real-world impact, efficiency, and business value.

TECHNICAL SKILLS

Machine Learning Clustering, Segmentation, Hyperparameter Tuning, Bagging, Boosting, Cross-Validation, Feature Engineering, Decision Trees, Random Forest, GBM, SVM, K-Means, KNN

Deep Learning & NLP RNN, CNN, Transformer Models, Transfer Learning, Tokenization, NER, Sentiment Analysis, Word Embeddings (Word2Vec)

Generative AI Lang Chain, Hugging Face, Open AIGPT, RAG, RLHF, PEFT, Intelligent Agents, Chatbots, Content Summarization

Cloud Computing AWS EC2, Lambda, S3, RDS, Dynamo DB, Azure Virtual Machines, Blob Storage, Cosmos DB, Cognitive Search

Programming Python, NumPy, Pandas, Scikit-learn, PyTorch, Flask, SQL, MySQL, PostgreSQL, Query Optimization, Data Warehousing

Tools & Platforms Git, GitHub, Jupyter Notebook, Google Colab, Tableau, Power BI

MLOps & Deployment Docker, Kubernetes, MLFlow, Apache Airflow, AWS Sage Maker

Prompt Engineering Few-shot/Zero-shot learning, Instruction tuning, Chain-of-thought prompting, Retrieval-augmented prompts for LLMs

EXPERIENCE & RESPONSIBILITIES

AI Enginner FORD	May 2024 - Present TX
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Responsibilities

- Designed and built a user-facing agentic GenAI system with deterministic routing across document-based RAG, LLM-driven text-to-SQL analytics, and constrained general knowledge queries.
- Implemented a production-grade RAG pipeline using Azure Document Intelligence, Azure OpenAI embeddings, and Azure AI Search with hybrid (vector + keyword) retrieval, enforcing citation-backed answers, retrieval confidence checks, and refusal/fallback behavior to prevent hallucinations.
- Developed an LLM-based text-to-SQL pipeline from scratch using schema injection and structured prompting, with post-generation SQL validation and execution guardrails including read-only access, table/column allowlists, row limits, and timeouts.
- Ensured strict grounding of structured data responses by transforming SQL query results into intermediate structured representations prior to LLM summarization, preventing model inference beyond database outputs.
- Built a state-based agent orchestration layer using LangChain and LangGraph to perform intent classification, tool eligibility checks, and controlled execution flow, preventing uncontrolled tool invocation.
- Implemented security and abuse-prevention mechanisms for a public-facing system, including prompt-injection defenses on user input and retrieved content, tool allowlisting, refusal logic, PII-safe filtering, rate limiting, and audit logging.
- Deployed backend services on Azure Kubernetes Service (AKS) using containerized workloads, designing stateless APIs with autoscaling, managed configuration, and persisted session state, and integrating Azure Monitor for latency tracking, failure analysis, and tool-level observability.
- Built a low-cost, AI-based conversation quality evaluation pipeline combining deterministic system signals (tool success, citations, latency, repeated queries) with lightweight LLM classification to label successful vs failed interactions and drive continuous improvement.

Environment

Python, LangChain, LangGraph, Azure OpenAI, Azure AI Search, Azure Document Intelligence, Azure SQL, Azure Kubernetes Service (AKS), Docker, Azure Monitor

ML Engineer

Designed and productionized an end-to-end NLP system for classifying customer sentiments on e-commerce reviews using TF-IDF, Logistic Regression, and SVM models, integrated into real-time workflows via a Flask REST API deployed on AWS EC2.

Responsibilities:

- Built a robust text preprocessing pipeline incorporating token normalization, lemmatization, stop-word filtering, and tokenization to standardize diverse user-generated content and improve model performance.
- Applied advanced NLP techniques including Named Entity Recognition (NER) and text classification to extract relevant features and entities from unstructured review data for better context understanding.
- Integrated semantic modeling using Word2Vec embeddings to capture nuanced meaning in customer language, enabling more accurate classification and feature extraction from textual data.
- Created scalable ETL pipelines using Apache Airflow and AWS Glue to automate ingestion, cleaning, and transformation of raw review datasets from various marketplaces and sources.
- Streamlined the machine learning lifecycle using MLflow for managing experiments, versioning models, and tracking metrics, enabling consistent deployment and reproducibility.
- Visualized insights such as sentiment trends, product feedback, and user behavior through dynamic dashboards in Tableau and Power BI, aiding marketing and product strategy teams in decision-making.
- Containerized NLP services with Docker and orchestrated deployment using Kubernetes to ensure high availability, fault tolerance, and easy scalability in production environments.
- Integrated performance monitoring and alerting using AWS CloudWatch to observe system health, model drift, and API latency, enabling proactive model tuning and maintenance.
- Collaborated with cross-functional teams including data analysts and backend developers to align NLP outputs with business KPIs, enhancing product insights and customer experience initiatives.

Environment: Python, Scikit-learn, Apache Airflow, AWS Glue, MLflow, Flask, TF-IDF, Logistic Regression, SVM, Word2Vec, Named Entity Recognition (NER), Docker, Kubernetes, AWS EC2, AWS CloudWatch, Tableau, Power BI.

SMART BRIDGE with IBM, India

Feb 2022– Apr 2022

Developer Intern

Designed and developed a CNN-based image classification model using the VGG16 architecture to detect common leaf diseases in fruits and vegetables, targeting early-stage agricultural intervention.

Responsibilities:

- Curated and cleaned a dataset of over 20,000 annotated leaf images spanning multiple plant species, applying preprocessing techniques such as resizing, normalization, and data augmentation to improve model generalization.
- Achieved high classification accuracy through fine-tuning of the VGG16 model using transfer learning, followed by validation with cross-entropy loss and confusion matrix analysis for performance benchmarking.
- Deployed the trained model as a Flask-based RESTful API, supporting real-time image predictions through HTTP endpoints accessible by external applications.
- Built a responsive and lightweight front-end web interface using HTML, CSS, and Bootstrap, enabling farmers and agritech users to upload leaf images and receive instant disease diagnosis results.
- Integrated OpenCV for image processing and validation at the front-end layer to enhance prediction quality and user feedback.
- Designed the system with modular components, enabling future integration with cloud platforms like AWS Lambda or Azure Functions for serverless scalability.
- Provided multilingual support and visual feedback features in the UI to improve usability for rural farming communities, focusing on accessibility and impact.
- Documented the model pipeline and deployment strategy, enabling easy handoff and future expansion by agricultural tech teams or academic researchers.
- Delivered a practical, AI-driven solution that supports smarter farming decisions, reduces crop loss, and promotes precision agriculture practices at scale.

Environment: Python, TensorFlow, Keras, VGG16, Flask, OpenCV, HTML, CSS, Bootstrap, NumPy, Pandas, REST API.